

ORCA: Open-ended Response Correctness Assessment for Audio Question Answering

Šimon Sedláček¹, Sara Barahona², Cecilia Bolaños³, Laura Herrera-Alarcón²,
Sathvik Udupa¹, Fernando López², Allison Ferner⁴, Bolaji Yusuf¹,
Alicia Lozano-Diez², Santosh Kesiraju¹, Ramani Duraiswami⁵, Jan Černocký¹

¹Speech@FIT, Brno University of Technology, Czechia ²AUDIAS, Universidad Autónoma de Madrid, Spain

³University of Buenos Aires, Argentina ⁴Tufts University, USA ⁵University of Maryland, USA

kesiraju@fit.vut.cz

Abstract

Reliable assessment of the abilities of large audio language models (LALMs) is essential to advancing the state of the art. As benchmarks rapidly evolve to incorporate complex reasoning and subjective tasks, they increasingly necessitate open-ended responses from LALMs. We present Open-ended Response Correctness Assessment (ORCA)—a reliable and lightweight model-based approach for answer correctness and disagreement modeling. We employ a three-stage annotation pipeline combining human judgment, structured feedback, and human-AI correction, yielding 9,663 annotations across 3,699 question-answer pairs from 15 LALMs on three audio understanding and reasoning benchmarks (achieving a Krippendorff’s alpha of 0.82). Our experiments employing curriculum learning show that ORCA models achieve a Spearman correlation of 0.91 with average human correctness ratings on seen benchmarks and generalize to unseen benchmarks with a score of 0.85, outperforming several LLM judge baselines including Gemini 2.5 Flash. Furthermore, we demonstrate that ORCA’s predicted variance correlates strongly with human disagreement, allowing it to effectively identify problematic benchmark items.

1 Introduction

Large audio language models (LALMs) are emerging as powerful tools for audio understanding, capable of answering questions about speech, music, and environmental sounds (Gemma Team, 2025b; Goel et al., 2025; Xu et al., 2025). To advance these models, efficient and reliable evaluation is essential. Most LALM evaluations use the multiple-choice question (MCQ) framework (Sakshi et al., 2024; Bhattacharya et al., 2025; Yang et al., 2025; Wang et al., 2026), as it enables fast and automatic assessment. Despite its

flaws (López et al., 2025), MCQ evaluation is convenient, but open-ended responses are more natural and better reflect real-world usage (Balepur et al., 2025). However, evaluating open-ended responses poses significant challenges including assignment of partial credits for incomplete answers, handling semantically equivalent answers, and uncertainty in human annotations.

Although classical lexical metrics such as F1 and BLEU are able to assign partial credit, the need to handle semantically equivalent answers makes them unsuitable (Bulian et al., 2022), and necessitates model-based alternatives (Kim et al., 2024; Yang et al., 2024) that leverage distributional semantics encoded in pretrained text encoders to handle answer equivalence.

Another important aspect of audio question answering (QA) is that human annotators often disagree on answer correctness. Even in tasks which are typically thought of as classification such as part-of-speech tagging and named entity recognition, annotators exhibit variance in judgment (Uma et al., 2022; Fornaciari et al., 2021). This is significantly exacerbated in Audio QA which entails subjective judgment about acoustic events, musical characteristics, emotional tone or sarcasm. The classical *source-channel* approach treats annotator disagreement as noise to be solved by aggregation and voting (Dawid and Skene, 1979; Hovy et al., 2013). We instead take the prescriptivist view that annotator disagreements can contain information about the genuine ambiguity in the questions and answers, and should be fully modeled rather than marginalized (Fleisig et al., 2024).

Figure 1 illustrates this phenomenon: two question-answer pairs receive similar mean ratings ($\bar{x} \approx 3.0$ on a 1–5 scale), yet one exhibits high annotator agreement ($s^2 = 0.4$) while the other shows genuine disagreement ($s^2 = 2.7$). To our knowledge, no prior work has modeled the full

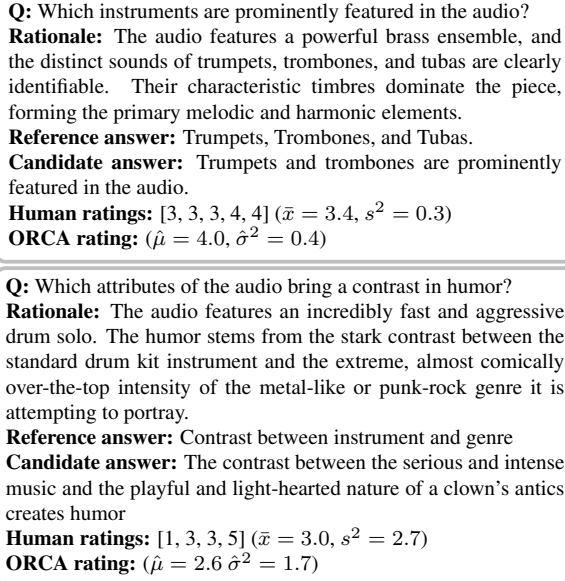


Figure 1: Two examples with similar mean ratings ($\bar{x} \approx 3$) but different variances: low variance (top) indicates consensus; high variance (bottom) reveals disagreement.

distribution of human correctness judgments for audio QA evaluation. Existing audio QA evaluation metrics fail to capture this distributional information, leaving them unable to reflect answers that would elicit disagreement among human annotators. Modeling such disagreement has practical benefits for end users and benchmark creators. For end users, it gives feedback about the confidence of a rating, especially for answers judged to be controversial. For benchmark creators, aggregating measures of disagreement could give insight into the quality of a benchmark and provide an early warning mechanism to trigger correction by human annotators; for instance, high variance across multiple candidate responses for a given question may indicate ambiguous framing.

We present ORCA (**O**pen-ended **R**esponse **C**orrectness **A**ssessment) for audio QA, a model-based framework which scores free-form answers from LALMs while modeling the variability in human judgments. For practical scalability, we adopt a text-only evaluation approach in which metrics assess the correctness of the answer augmented by textual grounding (rationales that summarize the pertinent parts of the audio context) (Yang et al., 2024), thereby avoiding the circular dependency of using audio models to judge audio models. Moreover, text-based models are far more efficient and are independent of the duration

of the audio. This evaluation framework applies uniformly to human annotators, ORCA, and LLM-judge baselines.

The contributions of this work are as follows.

- We propose ORCA, a lightweight model-based answer correctness assessment framework that predicts the distribution of human correctness judgments as continuous Beta or discrete categorical distributions, enabling robust uncertainty quantification for open-ended audio QA evaluation.
- We present a three-stage annotation framework that combines human judgment with structured feedback and refinement, yielding high-quality training and evaluation data while simultaneously improving the underlying benchmark quality.
- We collect 9,663 human annotations across 3,699 question-answer pairs from 15 state-of-the-art LALMs on three audio understanding and reasoning benchmarks (MMAU, MMAR, and MMAU-Pro), achieving a Krippendorff’s α above 0.81 post-filtering.
- We conduct extensive experiments using a three-stage curriculum learning approach that demonstrates the effectiveness of ORCA on both seen and unseen LALM responses and benchmarks. We show that ORCA outperforms several open-weight LLM-as-a-judge baselines and Gemini 2.5 Flash in predicting average annotator ratings at a fraction of the inference cost, while uniquely capturing annotation uncertainty.
- We release our trained models, source code, and curated annotation dataset to facilitate further research in this area¹.

2 Related work

Audio question answering benchmarks have evolved from foundational perception tasks to complex reasoning scenarios (Ma et al., 2025; Kumar et al., 2026), enabling us to evaluate large audio language models (LALMs) across diverse capabilities, which poses the following challenges.

¹<https://github.com/BUTSpeechFIT/ORCA>

2.1 Open-ended answer evaluation

As mentioned earlier, multiple-choice QA has been the most widely used evaluation strategy when benchmarking LALMs—since evaluating open-ended responses presents challenges. This is mainly due to semantic equivalence, partial correctness, and subjective interpretation. Traditional lexical matching (exact match, token-level F1, BLEU) does not capture semantic similarity (Bulian et al., 2022). Recent work explores LLM-based judges such as Prometheus (Kim et al., 2024) for text-based QA, while audio QA benchmarks such as AIR-Bench (Yang et al., 2024) and AudioBench (Wang et al., 2025) employ API-based LLM-judges with textual grounding (rationales). However, these approaches assume correctness of reference answers and rationales, whereas LLM-judges suffer from prompt sensitivity (Nalbandyan et al., 2025), lack of calibration, high computational costs, and reproducibility concerns (Chehbouni et al., 2025).

Our work addresses these limitations through our three-stage annotation framework and the flexibility of the proposed ORCA model. With the former, we systematically validate and refine textual grounding with structured feedback mechanisms and this process serves dual purposes: generating calibrated training data for ORCA while improving benchmark quality of LALMs.

2.2 Human annotation variability

Human annotation disagreement in subjective tasks often reflects genuine ambiguity rather than noise (Plank, 2022; Sandri et al., 2023; Fleisig et al., 2024). There is a rising trend of works which model rather than marginalize annotator disagreement in diverse text-based tasks including machine translation (Giovannotti, 2023), part-of-speech tagging (Fornaciari et al., 2021) and toxic speech detection (Mostafazadeh Davani et al., 2022; Leonardelli et al., 2023; Fleisig et al., 2023; Wu et al., 2024; Ni et al., 2026), as well as speech-based ones such as modeling listener bias for judging speech synthesis (Leng et al., 2021; Huang et al., 2022) and emotion recognition (Wu et al., 2023). However, no previous work has applied distributional modeling to assess answer correctness for open-ended QA, a gap which we address in this paper.

3 Annotation framework and datasets

Figure 2 illustrates our three-stage process for collecting high-quality human judgments for text-only evaluation while systematically improving benchmark quality.

3.1 Annotation pipeline

We prepare data through two parallel operations.

Stage 1a: Rationale generation To enable text-only evaluation models, we augment the benchmark data with textual grounding information (also referred to as *context*). A *rationale* is a textual justification explaining why the reference answer is correct given the audio and question. We generate rationales using Gemini-2.5-Flash (Gemini Team, 2025), prompting it with the audio, question, and reference answer. The rationale typically includes an audio description, reasoning steps, and when applicable, external knowledge. Gemini sometimes generates rationales based on textual cues rather than audio content, producing non-informative justifications. We detect these through human feedback in Stage 2, and corrections in Stage 3.

Stage 1b: Candidate answer generation We generate candidate answers by prompting multiple LALMs with each question and audio, creating a diverse set of responses for evaluation.

Stage 2: Annotation and feedback collection

Human annotators evaluate answer correctness given four pieces of textual information: the question, reference answer, rationale (includes transcript for speech questions), and a candidate answer. When textual context proves insufficient, annotators can listen to the original audio directly.

Annotators assign correctness scores on a 1–5 scale and provide structured feedback identifying issues through predefined categories: **Q** (incomplete question), **A** (insufficient rationale), **R** (incorrect reference), **U** (ambiguous instance), **E** (lacking expertise). Additional free-form comments capture issues not covered by these categories. This feedback enables targeted corrections in Stage 3. Lastly, LLM-judges perform parallel evaluations with the same textual inputs. Detailed rating definitions are given in Fig. 6 from Appendix A.

Stage 3: Corrections Humans review flagged instances and implement corrections: rephrasing

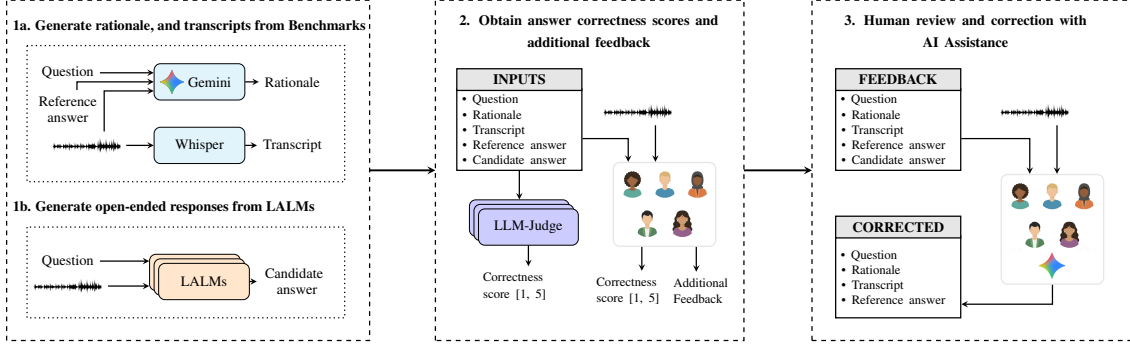


Figure 2: Annotation framework. Stage 1 generates rationales via Gemini, and candidate answers from LALMs using standard Benchmarks. Stage 2 collects correctness scores and structured feedback from humans and LLM-judges. Stage 3 (optional) implements human-AI corrections based on feedback.

questions (**Q**), enhancing rationales (**A**), and correcting reference answers (**R**). AI tools assist in generating improved content, but human experts validate all changes to ensure quality.

3.2 Benchmark set 1: MMAU & MMAR

The first phase of annotations was obtained on MMAU_{v05.15.25} (test-mini subset) (Sakshi et al., 2024) containing 1,000 questions spanning speech, sound, and music modalities, and MMAR (Ma et al., 2025) with 1,000 questions across speech, sound, music, and mixed-source modalities.

We generated candidate answers from 15 state-of-the-art audio language models namely Audio Flamingo 2 and 3 (Ghosh et al., 2025; Goel et al., 2025), Audio Reasoner (Zhifei et al., 2025), DeSTA2 & DeSTA2.5-Audio (Lu et al., 2025, 2026), GAMA (Ghosh et al., 2024), Gemma-3n (2B, 4B), GLM-4-Voice (Zeng et al., 2024), Kimi-Audio (Ding et al., 2025), Qwen2-Audio-7B & Qwen2-Audio-7B-Instruct, Qwen2.5-Omni-7B (Chu et al., 2024; Xu et al., 2025), and SALMONN (7B, 13B) (Tang et al., 2024). This yielded a total of 30,000 question-candidate answer pairs (2,000 questions $\times$ 15 models).

Out of 2,000 benchmark questions, 1,872 received at least one annotation (rating or feedback), leaving 128 questions unannotated. This coverage yielded 3,580 question-candidate answer pairs with correctness ratings (out of 30,000 possible pairs) and 1,993 feedback entries, totaling 11,721 annotations. Each rated pair received an average of 2.7 ratings. Details in Appendix A (Table 6).

Inter-annotator agreement on the 3,580 rated pairs, measured using Krippendorff’s alpha (Krippendorff, 2019), yielded $\alpha = 0.76$, indicating sub-

Field corrected	MMAU	MMAR	Total
Question (Q)	268	134	402 (20.1%)
Reference answer (R)	30	43	73 (3.6%)
Rationale (A)	166	150	316 (15.8%)
Total	464	327	791

Table 1: Number of fields corrected in Stage 3.

stantial agreement. However, the structured feedback revealed significant data quality issues: annotators flagged problems with questions (**Q**), reference answers (**R**), or rationales (**A**). Additionally, annotators reported when they were unable to judge due to ambiguity (**U**) or lack of expertise (**E**).

Following the annotation feedback, all flagged instances were reviewed over a two-week period, including the 128 questions that did not receive any annotations. Using the human-AI collaborative correction process described earlier, human annotators corrected problematic fields where necessary. Table 1 summarizes the number of corrections by field type and benchmark.

Based on the Stage 3 review, we filtered out annotations for 545 benchmark questions (out of 1,872 annotated). This invalidated 1,121 question-answer pairs (out of 3,580 rated pairs) and their associated 3,150 ratings (32% of total). Notably, these filtered ratings had substantially lower agreement ($\alpha = 0.59$), resulting from being unreliable instances rather than genuine disagreement. The remaining 6,578 valid ratings across 2,459 question-answer pairs showed improved agreement ($\alpha = 0.82$). Details in Appendix A (Fig. 7 and 8).

Benchmark	Questions	LALMs	QA pairs	Ratings	Avg/pair	Mean	α
MMAU _{test-mini}	619	15	1,144	3,056	2.67	2.82	0.82
MMAR	707	15	1,315	3,522	2.68	2.43	0.82
MMAU-Pro	248	8	1,240	3,085	2.49	2.35	0.83
Total	1,574		3,699	9,663			

Table 2: Annotation statistics for the three benchmarks after filtering. QA pairs = unique question–candidate answer pairs rated; Avg/pair = mean ratings per pair; Mean = mean correctness rating (1–5 scale); α = Krippendorff’s ordinal alpha (all three indicate strong agreement).

3.3 Benchmark set 2: MMAU-Pro

Building on lessons learned from the MMAU & MMAR annotation study, we refined both the rationale generation prompts and the annotation interface for MMAU-Pro (Kumar et al., 2026). Specifically, we developed a custom annotation UI that streamlined the annotator experience, and updated the Gemini-2.5-Flash prompting strategy to reduce non-informative rationales.

MMAU-Pro is a recently introduced benchmark comprising 5,305 audio QA questions across speech, sound, spatial audio, music and mixture modalities. We selected a stratified sample of 384 questions representative of the benchmark’s category and modality distribution. Candidate answers were generated from 8 LALMs selected from the original 15 to cover diverse model families and capability levels: Audio Flamingo 3, Audio Reasoner, DeSTA2.5-Audio, Gemma-3n (4B), Kimi-Audio, Qwen2-Audio-7B-Instruct, Qwen2.5-Omni-7B, and SALMONN (13B).

Unlike the MMAU & MMAR study, annotations for MMAU-Pro did not undergo Stage 3 correction. We obtained annotations for 1980 question-answer pairs. Instances with any annotator flags were excluded directly. Of the 384 questions, 95 carried at least one explicit quality flag: 92 (24%) were marked as unclear or requiring choices, 19 (2.3%) had a flawed reference answer, and 16 (4%) had an insufficient rationale; note that some questions carried multiple flags. A further 41 questions were skipped by all annotators without any rating or feedback. This left 248 clean, annotated questions spanning all categories and modalities of MMAU-Pro; in audio benchmarks, the primary source of variation is the audio content itself rather than question or answer type, making category-stratified coverage a more meaningful measure of representativeness than raw question count. These yielded 1,240 question-candidate answer pairs with 3,085 rat-

ings (average 2.5 ratings per pair). Inter-annotator agreement improved from $\alpha = 0.76$ on the full 384 question-answer pairs to $\alpha = 0.83$ after filtering, confirming that the excluded instances represented genuine annotation difficulty rather than random noise. Notably, only 4.2% of questions were flagged for insufficient rationales, compared to 15.8% in the MMAU & MMAR study, demonstrating the effectiveness of the refined prompting strategy. Details in Appendix A (Table 6).

Annotators Correctness ratings were collected from two groups. The first group comprised 37 volunteers — including professors, researchers, and graduate students from Jelinek Summer Workshop on Speech and Language Technology² who participated in a one-hour annotation session; instructions were explained in person prior to the task, and all participation was voluntary with the permission of the workshop organizers and the department head. The second group consisted of the authors—graduate students and researchers with backgrounds in speech and NLP who contributed additional ratings and performed all Stage 3 corrections and quality review.

Table 2 summarizes the final annotation statistics across all three benchmarks. All subsets achieve strong inter-annotator agreement ($\alpha > 0.81$) and are used in the experiments described in Section 5. A few examples of genuine annotator disagreements and statistics are given in Appendix A (Fig. 9 and Table 7).

4 The ORCA model

Open-ended answer evaluation presents a fundamental challenge: multiple human annotators independently grade on the correctness of the same candidate answer and sometimes disagree on the correctness. This can reflect genuine ambiguity rather than annotation noise. Let $\mathcal{D}$ represent a

²<https://jsalt2025.fit.vut.cz/>

dataset consisting of (question, answer, rationale, candidate) instances, where each instance i has N_i correctness ratings (integers) $\{y_{ij} \in \llbracket 1, 5 \rrbracket\}$ for $1 \leq j \leq N_i$.

4.1 Model architecture

ORCA is initialized from a pre-trained transformer-based *text* LLM. Given an instance consisting of a question q , reference answer r , rationale a , and candidate answer c , we concatenate these elements using separator tokens to obtain the final input: $x = [q; r; a; c]$. The concatenated input is tokenized and fed through the pre-trained LLM, producing contextualized representations. The last representation from the final hidden layer is extracted and transformed through a single linear layer to get an output vector: $\mathbf{z} = \mathbf{W} \mathbf{h}_{\text{final}} + \mathbf{b}$, whose dimensions are interpreted as the parameters of a likelihood function of the annotator ratings. We experiment with three such likelihood functions: Beta, Multinomial, and Bernoulli.

The Beta and Multinomial distributions model the full distribution of annotator ratings. The former directly imposes the inductive bias that ratings close to each other should be correlated while the latter treats each rating as an independent category, allowing the model to learn any cross-category correlations from the data itself. The Bernoulli formulation models only the average rating (but not the variance) and serves as a baseline that indicates how answer-correctness prediction is affected by uncertainty modeling. Thus, each objective induces a different dimensionality and interpretation of the vector $\mathbf{z}$, as will be further described in subsequent sections.

4.2 Beta

The Beta distribution is a natural choice for modeling the distribution of answer correctness ratings since it inherently respects the ordinal nature of correctness ratings through its parameterization. It is flexible enough to capture diverse rating patterns: high consensus (low variance) when most raters agree, high disagreement (high variance) when opinions diverge, and even U-shaped bimodal distributions when raters are polarized between considering an answer completely correct or completely incorrect. We scale the ratings from $\llbracket 1, 5 \rrbracket$ to $[0, 1]$ during training.

Here, the model output $\mathbf{z} \in \mathbb{R}^2$ represents $\log \alpha$ and $\log \beta$ to ensure that the Beta distribution pa-

rameters: $\alpha = \exp(\log \alpha)$ and $\beta = \exp(\log \beta)$ remain positive. These parameters define a Beta distribution on correctness scores in the range $[0, 1]$ with probability density function:

$$\text{Beta}(y; \alpha, \beta) = \frac{y^{\alpha-1}(1-y)^{\beta-1}}{\text{B}(\alpha, \beta)} \quad (1)$$

where $\text{B}(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$ is the Beta function. The expected correctness score and variance in $[1, 5]$ scale are directly obtained from the estimated parameters.

$$\hat{\mu}_{\text{Beta}} = 1 + \frac{4\alpha}{\alpha + \beta} \quad (2)$$

$$\hat{\sigma}_{\text{Beta}}^2 = \frac{16\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} \quad (3)$$

We frame the training as a maximum likelihood estimation problem. We treat each rating y_{ij} as an independent sample from the underlying Beta distribution.

The training objective is to maximize the log-likelihood:

$$\begin{aligned} \mathcal{L}_{\text{Beta}}(\theta) = \sum_{i \in \mathcal{D}} \left[(\alpha_i - 1) \sum_{j=1}^{N_i} \log(y_{ij}) \right. \\ \left. + (\beta_i - 1) \sum_{j=1}^{N_i} \log(1 - y_{ij}) - \log \text{B}(\alpha_i, \beta_i) \right] \quad (4) \end{aligned}$$

where α_i and β_i are the parameters predicted by the model for instance i , and θ represents all the trainable model parameters.

4.3 Multinomial

The Multinomial distribution models the likelihood of each discrete rating category independently, treating the $\llbracket 1, 5 \rrbracket$ Likert scale as five unordered classes. While this ignores the ordinal structure of the scale, the inter-category dependencies are captured implicitly through the training data. We normalize the $\mathbf{z} \in \mathbb{R}^5$ to $\phi = \text{softmax}(\mathbf{z})$. The reference ratings are mapped from the original $\llbracket 1, 5 \rrbracket$ scale to the discrete categories $y_{ij} \in \{1, 2, 3, 4, 5\}$ monotonically. We train the model to maximize the log-likelihood of the observed counts of the individual ratings.

$$\mathcal{L}_{\text{Multi}}(\theta) = \sum_{i \in \mathcal{D}} \sum_{j=1}^{N_i} y_{ij} \log \phi_{ij} \quad (5)$$

Here the expected correctness score and variance are $\hat{\mu}_{\text{Multi}} = \sum_{k=1}^5 k \cdot \phi_k$ and $\hat{\sigma}_{\text{Multi}}^2 = \sum_{k=1}^5 \phi_k (k - \hat{\mu}_{\text{Multi}})^2$ respectively.

4.4 Bernoulli

As a simpler alternative, we consider a Bernoulli baseline that models only the mean correctness without capturing the full distribution. In this variant, the ratings are normalized from the $\llbracket 1, 5 \rrbracket$ scale to $[0, 1]$, and $\bar{y}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} y_{ij}$ is their mean. The model predicts a single correctness value $\phi \in [0, 1]$ and is trained to maximize the log-likelihood under a Bernoulli distribution:

$$\mathcal{L}_{\text{Bern}}(\theta) = \sum_{i \in \mathcal{D}} \bar{y}_i \log \phi_i + (1 - \bar{y}_i) \log(1 - \phi_i) \quad (6)$$

The expected correctness score on the $[1, 5]$ scale is given by $\hat{\mu}_{\text{Bern}} = 1 + 4\phi$, but this model does not provide a variance estimate as it does not capture the distribution of ratings.

5 Experiments

5.1 Evaluation protocol

We design two evaluation scenarios along two axes—question familiarity and LALM familiarity—each evaluated on two test sets (Table 3).

Scenario A: Unseen questions, seen LALMs

We conduct five-fold cross-validation experiments using the 2,459 valid question-answer pairs from MMAU and MMAR: five independent train/dev/test splits (8:1:1 ratio), stratified by audio modality (speech, sound, music) and question category (perceptual skill). Test questions have no overlap with training questions, while all 15 LALMs appear in both training and test sets, yielding approximately 266 test QA pairs per split. The same 5 trained models are additionally evaluated on the full MMAU-Pro test set (1,240 pairs, seen LALMs). We report mean $\pm$ std across the 5 runs.

Scenario B: Unseen LALMs We construct four splits each holding out 2 LALMs (covering 8 LALMs in total across four splits), training on the remaining 13 LALMs from MMAU and MMAR. Test responses come from the held-out LALMs, yielding approximately 440 test QA pairs per split. The same 4 trained models are additionally evaluated on the corresponding MMAU-Pro subset (310 pairs from the held-out LALMs). We report mean $\pm$ std across the 4 splits.

5.2 Training data

ORCA is trained using a three-stage curriculum, with each stage providing training signal of in-

Scenario	Splits	MMAU & MMAR	MMAU-Pro
A	5	266	1,240 [†]
B	4	440	310

Table 3: Evaluation scenarios. Numbers indicate average number of test QA pairs per split. ([†]MMAU-Pro is a fixed test set evaluated across all 5 runs of Scenario A.) MMAU-Pro is never used in training.

creasing quality and specificity. Across all stages, instances where any text exceeds 1,000 characters are removed; reported instance counts are after this filtering.

Stage 1 (s1): Synthetic LLM-judge data

We use the publicly available training corpus from Audio Flamingo 3, covering speech, sound, and music QA pairs. We prompted 5 LLMs—Gemma 3 (4B, 12B) (Gemma Team, 2025a), Llama 3.1-8B (Llama Team, 2024), Qwen2.5-7B-Instruct, and Qwen3-14B (Qwen Team et al., 2025)—to generate synthetic candidate answers at varying correctness levels (correct, partially correct, incorrect) and then to rate each pair, yielding 5 million instances (Prompts in Appendix B Figs. 10 and 11). This stage does not include rationales.

Stage 2 (s2): LLM-judge data on seen benchmarks

Stage 2 uses the 2,000 questions from MMAU_{test-mini} and MMAR (post correction from Section 3.2), augmented with rationales as described in Section 3. Candidate answers from the 15 LALMs yield 30,000 question-candidate answer pairs. We additionally prompt the same 5 LLMs to generate paraphrased variants of each existing candidate answer at varying correctness levels, producing approximately 450k additional pairs. All 480k instances are rated by the same 5 LLMs; unlike s1, this stage includes rationales. To prevent test data leakage, s2 instances are further filtered per evaluation split: for Scenario A splits, instances whose questions appear in the held-out test set are removed (414k remaining); for Scenario B splits, instances from the 2 held-out LALMs and their paraphrases are removed (410k remaining).

Stage 3 (s3): Human-annotated data

Stage 3 uses the human-annotated MMAU and MMAR data and splits described in Section 5.1, ensuring no leakage between training and test data. Each split provides approximately 1,960 training pairs with their individual human ratings. MMAU-Pro

is never used for training and serves exclusively as an unseen benchmark test set across all the experiments.

5.3 Evaluation metrics

All predicted scores are evaluated on the $[1, 5]$ scale. For ranking quality, we compute Spearman’s ρ and Kendall’s τ rank correlation coefficients between predicted and human mean correctness scores. We report mean absolute error for expected correctness (MAE_μ) to evaluate correctness prediction for all models. For ORCA models, we additionally compute mean absolute error for variance (MAE_{σ^2}) to evaluate variance prediction.

5.4 ORCA training

We fine-tune OLMo-2 (1B) (Walsh et al., 2025), Gemma 3 (1B, 4B), and Llama 3.2 (1B, 3B) with LoRA ($r = 128$, $\alpha = r$) (Hu et al., 2022).

We follow a curriculum training approach systematically progressing through stages $s1 \rightarrow s2 \rightarrow s3$, with maximum training steps of 10k, 5k and 2k, and peak learning rates of $5e-5$, $2e-5$ and $2e-5$, respectively. We use a linear warmup followed by linear decay schedule: the learning rate rises linearly from 0 to the peak over the warmup steps, then decays linearly to a minimum floor of $0.1 \times \text{peak LR}$. We maintain an effective batch size of 16 and use early stopping with a patience of 5 based on Spearman ρ on the respective dev set. On a single 48GB NVIDIA A6000 GPU, the full curriculum training for OLMo2-1B with LoRA 128 is completed in about 5 hours.

Post-processing Unlike humans and LLM-judges, the Beta, Multinomial, and Bernoulli objectives cannot naturally predict hard 1s and 5s. This creates noisy predictions at the extremes. Noting that, clear 1 and 5 ratings typically have low variance for both humans and ORCA, we devise a simple *clamping* scheme. For Beta and Multinomial, we set the predicted score to 1 or 5 if the original prediction is within 0.5 of the respective extreme and the predicted variance falls below a threshold optimized on the development set to maximize $\rho + \tau$. For Bernoulli, which does not predict variance, we apply score-only clamping: predictions within 0.5 of 1 or 5 are set to the respective extreme.

5.5 LLM-judge baselines

We evaluate the following judges across multiple prompting conditions of increasing contextual richness; the full prompting ablation and prompt templates are provided in Appendix B and C (Table 9).

Offline open-weight LLMs Llama 3.1-8B, Qwen2.5-7B, and Gemma 3 (4B, 12B).

Prometheus We include Prometheus 2-7B, an open-weight LLM explicitly fine-tuned for evaluation tasks, which rates responses based on rubrics and reference answers.

API-based LLM-judge We also evaluate Gemini-2.5-Flash, a proprietary API-based model.

6 Results and analysis

6.1 Comparison of ORCA with LLM-judges

In Table 4, ORCA is compared with five offline LLM-judges and Gemini-2.5-Flash. The table presents results for six ORCA models based on the OLMo2-1B and Llama3.2-3B architectures, all trained using the full $s1 \rightarrow s2 \rightarrow s3$ curriculum. We train each base model once with each of the three objectives: Bernoulli, Beta, Multinomial. We follow evaluation scenario A (see Section 5.1), reporting the mean and standard deviation for each metric.

Both the Multinomial and Bernoulli Llama models outperform all LLM-judges, with the Bernoulli model leading slightly ($\rho = 84.89$ on the MMAU-Pro test set). While ORCAs are best compared to the Gemma3-4B LLM-judge given their sizes, they surpass even Gemini-2.5-Flash in human correlation and MAE (7.76 vs. 8.90). Moreover, OLMo2-1B ORCA models outperform the much larger Gemma3-12B and come very close to Gemini. This highlights the viability and efficiency of smaller ORCA models, which require only a single forward pass for the final score prediction. ORCA shows strong generalization to novel benchmarks, outperforming the reference judges on unseen MMAU-Pro questions. Section 6.2 further details its generalization to unseen LLM responses. Finally, we compare the prediction bias (relative to the mean human score) of ORCA (Llama3.2-3B Multi.) against Gemini-2.5-Flash in Figure 13 (Appendix C).

Model	Objective	MMAU/MMAR test			MMAU-Pro test		
		$\tau (\times 100)$	$\rho (\times 100)$	MAE (%)	$\tau (\times 100)$	$\rho (\times 100)$	MAE (%)
ORCA (Stage 3 — $s1 \rightarrow s2 \rightarrow s3$ curriculum)							
OLMo2-1B	Beta	78.53 $\pm$ 1.15	89.65 $\pm$ 1.06	8.68 $\pm$ 0.51	72.47 $\pm$ 2.14	82.74 $\pm$ 0.97	9.35 $\pm$ 0.77
	Multinomial	78.32 $\pm$ 1.81	89.14 $\pm$ 1.03	8.56 $\pm$ 0.10	73.18 $\pm$ 3.10	83.18 $\pm$ 1.12	8.60 $\pm$ 0.46
	Bernoulli	79.47 $\pm$ 0.67	89.95 $\pm$ 0.40	8.51 $\pm$ 0.51	74.62 $\pm$ 0.63	83.66 $\pm$ 0.77	8.50 $\pm$ 0.15
Llama3.2-3B	Beta	79.66 $\pm$ 1.85	90.87 $\pm$ 0.99	7.95 $\pm$ 0.91	71.80 $\pm$ 3.48	82.78 $\pm$ 1.94	8.61 $\pm$ 1.09
	Multinomial	81.24 $\pm$ 0.91	91.22 $\pm$ 0.63	7.61 $\pm$ 0.61	75.43 $\pm$ 0.90	84.51 $\pm$ 0.60	7.79 $\pm$ 0.34
	Bernoulli	81.45 $\pm$ 0.91	91.20 $\pm$ 0.74	7.65 $\pm$ 0.81	76.06 $\pm$ 0.66	84.89 $\pm$ 0.25	7.76 $\pm$ 0.25
LLM-judges							
Gemini-2.5-Flash		80.70 $\pm$ 0.81	89.98 $\pm$ 0.73	9.11 $\pm$ 0.54	75.82	83.89	8.90
Gemma3-12B		79.89 $\pm$ 0.81	89.27 $\pm$ 0.45	11.09 $\pm$ 0.57	72.79	82.06	11.54
Gemma3-4B		70.95 $\pm$ 0.51	81.49 $\pm$ 0.52	14.71 $\pm$ 0.57	60.15	70.17	17.90
Llama3.1-8B		73.25 $\pm$ 1.32	84.35 $\pm$ 1.14	12.79 $\pm$ 0.67	66.85	76.38	12.81
Qwen2.5-7B		75.14 $\pm$ 0.86	85.53 $\pm$ 0.76	12.05 $\pm$ 0.28	68.69	78.14	12.13
Prometheus2-7B		63.62 $\pm$ 1.42	74.39 $\pm$ 1.60	19.41 $\pm$ 0.74	45.85	54.36	23.32

Table 4: Performance comparison of fine-tuned models and LLM-judges on the *MMAU/MMAR* and *MMAU-Pro* test splits. τ (Kendall) and ρ (Spearman) are scaled by 100 for better readability. Additional ORCA models are shown in Appendix C in Table 8.

These results indicate that the Multinomial is the most attractive of the three ORCA variants. Of the two variants with disagreement modeling, it consistently outperforms the Beta formulation, especially in out-of-domain experiments. Furthermore, although the Bernoulli variant is slightly better on average at predicting *average* answer correctness rating, the Multinomial variant performs almost as well—with correlations and MAE within the empirical standard deviations (across multiple random seeds) of the corresponding Bernoulli results—while also providing a measure of annotator disagreement.

6.2 Generalization to unseen models

To further test the generalization of ORCA, we compare Llama3.2-3B (Multi.) with Gemini-2.5-Flash, training and evaluating for scenario B as described in Section 5.1. This way, we ablate ORCA’s generalization to LALMs that were previously unseen during training. As shown in Figure 3, ORCA outperforms Gemini on most unseen LALMs, notably on AudioReasoner, whose responses generally have the highest variance in human ratings, as it is the most verbose and distinct in response style out of all LALMs used in this work.

6.3 Effect of curriculum learning

Table 5 shows the effects of pre-training using purely synthetic data in stages $s1$ and $s2$. We use Llama3.2-3B (Multi.) for this ablation and

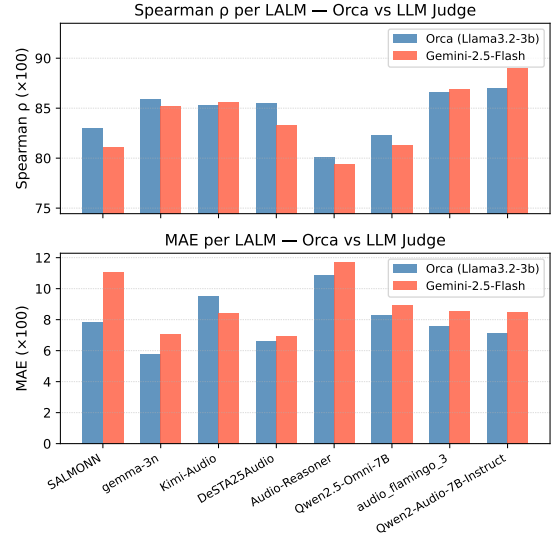


Figure 3: ORCA performance on LALMs unseen during training, comparing to Gemini-2.5-Flash on MMAU-Pro.

evaluate performance on unseen questions from MMAU-Pro. We observe that pre-training on $s1$ and $s2$ consistently improves performance after subsequent fine-tuning on human-annotated data ($s3$) and is better than standalone $s3$ training. While $s1 \rightarrow s3$ brings an improvement of 0.59 ρ to $s3$, it is less pronounced compared to the improvement brought by pretraining on $s2$ (+1.36 over $s3$), as $s2$ is already constructed from benchmark data, only synthetically rated by LLM-judges and inflated by augmentations like answer rephrasing.

Overall, $s1 \rightarrow s2 \rightarrow s3$ proved to be the best training strategy.

Staging strategy	$\rho (\times 100)$	MAE (%)
$s1$	73.26 ± 0.00	22.50 ± 0.00
$s2$	80.65 ± 0.86	10.46 ± 0.71
$s1 \rightarrow s2$	80.15 ± 0.83	10.60 ± 0.67
$s3$	82.96 ± 0.83	8.72 ± 0.30
$s1 \rightarrow s3$	83.55 ± 0.72	8.32 ± 0.21
$s2 \rightarrow s3$	84.32 ± 1.19	7.89 ± 0.41
$s1 \rightarrow s2 \rightarrow s3$	84.51 ± 0.60	7.79 ± 0.34

Table 5: Effect of pre-training strategy on ORCA Llama3.2-3B (Multi.) performance on MMAU-Pro.

6.4 Rationale ablations

We ablate the effect of using rationales in the input of both ORCA and LLM-judges. For the comparison (see Figure 4), we use Llama3.2-3B (Multi.) trained in the $s1 \rightarrow s2 \rightarrow s3$ scenario A setting, and compare it against the two strongest LLM-judges: Gemini-2.5-Flash and Gemma3-12B. While for ORCA the effect is less pronounced (presumably since ORCA models receive task-specific training), the rationale has a clear positive effect on the Spearman’s ρ as well as MAE. For LLM-judges, the improvements obtained with rationales underline the importance of supplying additional context relevant to the question, supporting the hypothesis that the original audio can be replaced with a textual approximation for the purposes of answer correctness evaluation.

6.5 Applications of predicted uncertainty

Correlation with human disagreement To validate that ORCA variance reflects genuine annotation difficulty, we compute Spearman ρ and MAE_{σ^2} between ORCA predicted variance and human inter-annotator variance for each of the eight held-out LALMs (items with ≥ 3 annotations from Scenario B). The correlation is significant for 7 of 8 LALMs ($0.22 < \rho < 0.74$; see Appendix C), confirming that ORCA reliably identifies responses where human raters genuinely disagree.

Correlation with flagged instances From Figure 5, ORCA-predicted variance is significantly higher for items flagged by human reviewers as ambiguous or problematic compared to non-flagged (clean) items. The figure shows Gaussian

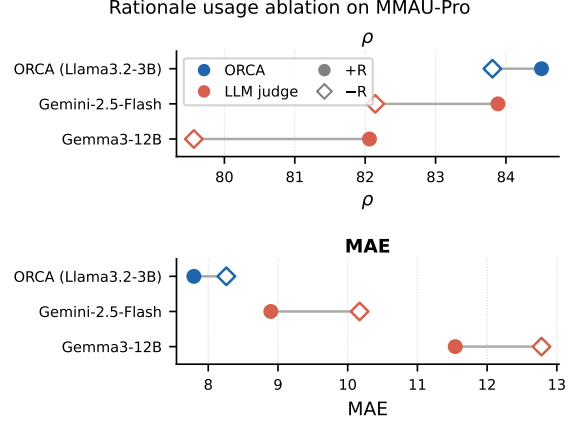


Figure 4: Effect of input rationale on ORCA Llama3.2-3B (Multi.) and LLM-judge performance on MMAU-Pro. (+R) and (-R) denote the inclusion/omission of rationale in the input, respectively.

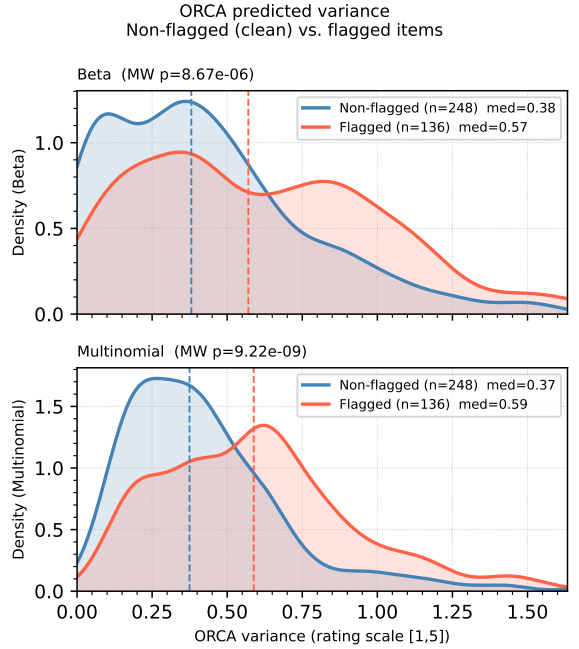


Figure 5: Gaussian kernel density estimate illustrating the distribution of predicted ORCA variances for the human annotated examples. Flagged instances are shown in red, while clean instances are shown in blue. Dashed lines indicate median.

kernel density estimation (KDE) of per-question ORCA variance (on the 1–5 rating scale) for 248 clean and 136 flagged questions from MMAU-Pro, evaluated with two parameterizations of the scoring model: Multinomial and Beta. In both cases the flagged distribution is stochastically dominant over the clean distribution (Mann-Whitney $p < 0.0001$), with median variance shifting from 0.38 to 0.59 (Multi.) and from 0.38 to 0.57 (Beta). This

indicates that ORCA assigns higher predictive uncertainty to questions that human reviewers independently identified as having annotation-quality issues, without having access to those review labels at inference time. See Appendix C for variance distributions for specific question flag categories.

7 Conclusions

We presented ORCA, a reliable and lightweight model-based approach for open-ended response correctness assessment in audio question answering. Our comparative analysis of three objective functions for modeling answer correctness showed that although a Bernoulli objective achieves the highest absolute Spearman correlation for expected correctness, it is inherently limited by its inability to model human disagreement. In contrast, the Multinomial distribution provides the most robust and practical framework, matching the correctness performance of Bernoulli within its variance range while effectively capturing human disagreement better than Beta. Our three-stage curriculum learning paradigm demonstrates that these models successfully leverage large-scale synthetic data and LLM-judge ratings, reducing the dependency on extensive human-annotated datasets. Furthermore, evaluation via mean absolute error demonstrated that ORCA’s predictions are highly calibrated, significantly outperforming LLM-judge baselines in absolute accuracy, while offering substantially greater computational efficiency over LLM alternatives by requiring only a single forward pass.

Beyond evaluation metrics, our pipeline exposed some data quality issues in existing datasets, allowing us to release corrected versions of these benchmarks. We demonstrated that ORCA’s predicted variance can serve a dual utility: estimating human disagreement on inherently subjective questions, and diagnosing underlying quality issues in benchmark items. While the continuous Beta distribution yielded slightly lower correlation scores in our experiments, we anticipate its practical utility will manifest when scaling to integrate heterogeneous rating scales across disparate datasets. Ultimately, this work serves as a foundational stepping stone toward shifting audio QA evaluation away from restrictive multiple-choice paradigms toward natural, real-world open-ended assessments.

Author contributions

ORCA model: Bolaji Yusuf, Santosh Kesiraju, Šimon Sedláček

LLM-as-a-judge: Sara Barahona, Laura Herrera-Alarcón, Cecilia Bolaños, Alicia Lozano-Diez

Analysis of existing benchmarks: Fernando López, Allison Ferner, Sara Barahona, Laura Herrera-Alarcón, Cecilia Bolaños, Alicia Lozano-Diez

LALM inference and analysis: Sathvik Udupa

ORCA experiments: Bolaji Yusuf, Šimon Sedláček, Santosh Kesiraju

User interface: Santosh Kesiraju — all authors provided useful feedback.

Project support and general guidance: Alicia Lozano-Diez, Ramani Duraiswami, Santosh Kesiraju, Jan Černocký

Writing: Šimon Sedláček, Laura Herrera-Alarcón, Sara Barahona, Alicia Lozano-Diez, Bolaji Yusuf, Santosh Kesiraju

Annotations: All authors and volunteers from JSALT’25.

Acknowledgments

The authors would like to thank Prof. Sanjeev Khudanpur and Prof. Jordan L. Boyd-Graber for several useful discussions and constructive feedback during the early stages of the work. We would like to thank the action editor and the reviewers for their invaluable guidance. We are especially grateful for the reviewers’ emphasis on rigorous out-of-domain evaluation, which inspired our curriculum learning framework and greatly enhanced the quality of this work.

Funding. Part of the work was done during JSALT’2025 Workshop which was supported by gifts from Johns Hopkins University, Google, and Phonexia. Ramani Duraiswami’s work was supported by ONR Award N00014-23-1-2086. This work was supported by European Union’s Horizon 2020 research and innovation programme under the Marie Skłodowska-Curie grant agreement No. 101007666; FPI PRE2022-104808 and PREP2024-003414 both funded by MICIU/AEI/10.13039/501100011033 and FSE+, project PID2024-160789OB-I00 funded by MICIU/AEI/10.13039/501100011033/FEDER, UE and project SI4/PJI/2024-00237 (COSER-IA), Comunidad de Madrid; Czech Ministry of Culture NAKI III project JARIN (DH23P03OVV010) and by Ministry of Education (MoE), Youth

and Sports of the Czech Republic through the OP JAK project “Linguistics, Artificial Intelligence and Language and Speech Technologies: from Research to Applications” (ID:CZ.02.01.01/00/23_020/0008518). Computing on IT4I supercomputer was supported by MoE through the e-INFRA CZ (ID:90254).

References

- Nishant Balepur, Rachel Rudinger, and Jordan Lee Boyd-Graber. 2025. [Which of these best describes multiple choice evaluation with LLMs? a\) forced B\) flawed C\) fixable D\) all of the above](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3394–3418, Vienna, Austria. Association for Computational Linguistics.
- Debarpan Bhattacharya, Apoorva Kulkarni, and Sriram Ganapathy. 2025. [Benchmarking and Confidence Evaluation of LALMs For Temporal Reasoning](#). In *Interspeech 2025*, pages 2068–2072.
- Jannis Bulian, Christian Buck, Wojciech Gajewski, Benjamin Börschinger, and Tal Schuster. 2022. [Tomayto, tomahto. beyond token-level answer equivalence for question answering evaluation](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 291–305, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Khaoula Chehbouni, Mohammed Haddou, Jackie CK Cheung, and Golnoosh Farnadi. 2025. [Neither Valid nor Reliable? Investigating the Use of LLMs as Judges](#). In *The Thirty-Ninth Annual Conference on Neural Information Processing Systems Position Paper Track*.
- Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, Chang Zhou, and Jingren Zhou. 2024. [Qwen2-Audio Technical Report](#).
- Alexander P. Dawid and Allan M. Skene. 1979. [Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm](#). *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 28(1):20–28.
- Ding Ding, Zeqian Ju, and Yichong Leng. 2025. [Kimi-Audio Technical Report](#). Technical report, MoonshotAI.
- Eve Fleisig, Rediet Abebe, and Dan Klein. 2023. [When the majority is wrong: Modeling annotator disagreement for subjective tasks](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 6715–6726, Singapore. Association for Computational Linguistics.
- Eve Fleisig, Su Lin Blodgett, Dan Klein, and Zeerak Talat. 2024. [The perspectivist paradigm shift: Assumptions and challenges of capturing human labels](#). In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2279–2292, Mexico City, Mexico. Association for Computational Linguistics.
- Tommaso Fornaciari, Alexandra Uma, Silviu Paun, Barbara Plank, Dirk Hovy, and Massimo Poesio. 2021. [Beyond black & white: Leveraging annotator disagreement via soft-label multi-task learning](#). In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 2591–2597, Online. Association for Computational Linguistics.
- Gemini Team. 2025. [Gemini 2.5: Pushing the Frontier with Advanced Reasoning, Multimodality, Long Context, and Next Generation Agentic Capabilities](#).
- Gemma Team. 2025a. [Gemma 3 Technical Report](#).
- Gemma Team. 2025b. [Gemma 3n](#).
- Sreyan Ghosh, Zhifeng Kong, Sonal Kumar, S. Sakshi, Jaehyeon Kim, Wei Ping, Rafael Valle, Dinesh Manocha, and Bryan Catanzaro. 2025. [Audio Flamingo 2: An Audio-Language Model with Long-Audio Understanding and Expert Reasoning Abilities](#). In *Proceedings of the 42nd International Conference on Machine Learning*. PMLR.

- Sreyan Ghosh, Sonal Kumar, Ashish Seth, Chandra Kiran Reddy Evuru, Utkarsh Tyagi, S Sakshi, Oriol Nieto, Ramani Duraiswami, and Dinesh Manocha. 2024. [GAMA: A large audio-language model with advanced audio understanding and complex reasoning abilities](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6288–6313, Miami, Florida, USA. Association for Computational Linguistics.
- Patrizio Giovannotti. 2023. [Evaluating Machine Translation Quality with Conformal Predictive Distributions](#). In *Proceedings of the Twelfth Symposium on Conformal and Probabilistic Prediction with Applications*, volume 204 of *Proceedings of Machine Learning Research*, pages 413–429. PMLR.
- Arushi Goel, Sreyan Ghosh, Jaehyeon Kim, Sonal Kumar, Zhifeng Kong, Sang gil Lee, Chao-Han Huck Yang, Ramani Duraiswami, Dinesh Manocha, Rafael Valle, and Bryan Catanzaro. 2025. [Audio Flamingo 3: Advancing Audio Intelligence with Fully Open Large Audio Language Models](#). In *Advances in Neural Information Processing Systems*, volume 38, pages 41819–41886. Curran Associates, Inc.
- Dirk Hovy, Taylor Berg-Kirkpatrick, Ashish Vaswani, and Eduard Hovy. 2013. [Learning whom to trust with MACE](#). In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1120–1130, Atlanta, Georgia. Association for Computational Linguistics.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. [LoRA: Low-Rank Adaptation of Large Language Models](#). In *The Tenth International Conference on Learning Representations, ICLR*. OpenReview.net.
- Wen-Chin Huang, Erica Cooper, Junichi Yamagishi, and Tomoki Toda. 2022. [LDNet: Unified Listener Dependent Modeling in MOS Prediction for Synthetic Speech](#). In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 896–900. IEEE.
- Seungone Kim, Juyoung Suk, Shayne Longpre, Bill Yuchen Lin, Jamin Shin, Sean Welleck, Graham Neubig, Moontae Lee, Kyungjae Lee, and Minjoon Seo. 2024. [Prometheus 2: An open source language model specialized in evaluating other language models](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 4334–4353, Miami, Florida, USA. Association for Computational Linguistics.
- Klaus Krippendorff. 2019. [Content Analysis: An Introduction to Its Methodology](#). SAGE Publications, Inc.
- Sonal Kumar, Šimon Sedláček, Vaibhavi Lokegaonkar, Fernando López, Wenyi Yu, Nishit Anand, Hyeonggon Ryu, Lichang Chen, Maxim Plička, Miroslav Hlaváček, William Fineas Ellingwood, Sathvik Udupa, Siyuan Hou, Allison Ferner, Sara Barahona, Cecilia Bolaños, Satish Rahi, Laura Herrera-Alarcón, Satvik Dixit, Rupali S. Patil, Soham Deshmukh, Lasha Koroshinadze, Yao Liu, Leibny Paola García-Perera, Eleni Zanou, Themis Stafylakis, Joon Son Chung, David Harwath, Chao Zhang, Dinesh Manocha, Alicia Lozano-Diez, Santosh Kesiraju, Sreyan Ghosh, and Ramani Duraiswami. 2026. [MMAU-Pro: A Challenging and Comprehensive Benchmark for Holistic Evaluation of Audio General Intelligence](#). In *Fortieth AAAI Conference on Artificial Intelligence, Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence, Sixteenth Symposium on Educational Advances in Artificial Intelligence, AAAI*, pages 22688–22697, Singapore. AAAI Press.
- Yichong Leng, Xu Tan, Sheng Zhao, Frank Soong, Xiang-Yang Li, and Tao Qin. 2021. [MBNET: MOS Prediction for Synthesized Speech with Mean-Bias Network](#). In *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 391–395. IEEE.
- Elisa Leonardelli, Gavin Abercrombie, Dina Al-maney, Valerio Basile, Tommaso Fornaciari, Barbara Plank, Verena Rieser, Alexandra Uma, and Massimo Poesio. 2023. [SemEval-2023 task 11: Learning with disagreements \(LeWiDi\)](#). In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, pages

- 2304–2318, Toronto, Canada. Association for Computational Linguistics.
- Llama Team. 2024. [The Llama 3 Herd of Models](#).
- Ke-Han Lu, Zhehuai Chen, Szu-Wei Fu, Chao-Han Huck Yang, Jagadeesh Balam, Boris Ginsburg, Yu-Chiang Frank Wang, and Hung-Yi Lee. 2025. [Developing Instruction-Following Speech Language Model Without Speech Instruction-Tuning Data](#). In *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5.
- Ke-Han Lu, Zhehuai Chen, Szu-Wei Fu, Chao-Han Huck Yang, Sung-Feng Huang, Chih-Kai Yang, Chee-En Yu, Chun-Wei Chen, Wei-Chih Chen, Chien-yu Huang, Yi-Cheng Lin, Yu-Xiang Lin, Chi-An Fu, Chun-Yi Kuan, Wenze Ren, Xuanjun Chen, Wei-Ping Huang, En-Pei Hu, Tzu-Quan Lin, Yuan-Kuei Wu, Kuan-Po Huang, Hsiao-Ying Huang, Huang-Cheng Chou, Kai-Wei Chang, Cheng-Han Chiang, Boris Ginsburg, Yu-Chiang Frank Wang, and Hung-yi Lee. 2026. [DeSTA2.5-Audio: Toward General-Purpose Large Audio Language Model With Self-Generated Cross-Modal Alignment](#). *IEEE Transactions on Audio, Speech and Language Processing*, 34:2062–2076.
- Fernando López, Santosh Kesiraju, and Jordi Luque. 2025. [Robustness assessment of large audio language models in multiple-choice evaluation](#).
- Ziyang Ma, Yinghao Ma, Yanqiao Zhu, Chen Yang, Yi-Wen Chao, Ruiyang Xu, Wenxi Chen, Yuanzhe Chen, Zhuo Chen, Jian Cong, Kai Li, Kelian Li, Siyou Li, Xinfeng Li, Xiquan Li, Zheng Lian, Yuzhe Liang, Minghao Liu, Zhikang Niu, Tianrui Wang, Yuping Wang, Yuxuan Wang, Yihao Wu, Guanrou Yang, Jianwei Yu, Ruibin Yuan, Zhisheng Zheng, Ziya Zhou, Haina Zhu, Wei Xue, Emmanouil Benetos, Kai Yu, EngSiong Chng, and Xie Chen. 2025. [MMAR: A Challenging Benchmark for Deep Reasoning in Speech, Audio, Music, and Their Mix](#). In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*.
- Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. 2022. [Dealing with disagreements: Looking beyond the majority vote in subjective annotations](#). *Transactions of the Association for Computational Linguistics*, 10:92–110.
- Grigor Nalbandyan, Rima Shahbazyan, and Evelina Bakhturina. 2025. [SCORE: Systematic Consistency and robustness evaluation for large language models](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 3: Industry Track)*, pages 470–484, Albuquerque, New Mexico. Association for Computational Linguistics.
- Jingwei Ni, Yu Fan, Vilém Zouhar, Donya Rooein, Alexander Miserlis Hoyle, Mrinmaya Sachan, Markus Leippold, Dirk Hovy, and Elliott Ash. 2026. [Can reasoning help large language models capture human annotator disagreement?](#) In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 36–54, Rabat, Morocco. Association for Computational Linguistics.
- Barbara Plank. 2022. [The “problem” of human label variation: On ground truth in data, modeling and evaluation](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 10671–10682, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.
- Qwen Team, An Yang, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chengyuan Li, Dayiheng Liu, Fei Huang, Haoran Wei, Huan Lin, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jingren Zhou, Junyang Lin, Kai Dang, Keming Lu, Keqin Bao, Kexin Yang, Le Yu, Mei Li, Mingfeng Xue, Pei Zhang, Qin Zhu, Rui Men, Runji Lin, Tianhao Li, Tianyi Tang, Tingyu Xia, Xingzhang Ren, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yu Wan, Yuqiong Liu, Zeyu Cui, Zhenru Zhang, and Zihan Qiu. 2025. [Qwen2.5 Technical Report](#).
- S. Sakshi, Utkarsh Tyagi, Sonal Kumar, Ashish Seth, Ramaneswaran Selvakumar, Oriol Nieto, Ramani Duraiswami, Sreyan Ghosh, and Dinesh Manocha. 2024. [MMAU: A Massive Multi-Task Audio Understanding and Reason-](#)

- ing Benchmark. In *The Twelfth International Conference on Learning Representations*.
- Marta Sandri, Elisa Leonardelli, Sara Tonelli, and Elisabetta Jezek. 2023. [Why don't you do it right? analysing annotators' disagreement in subjective tasks](#). In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics*, pages 2428–2441, Dubrovnik, Croatia. Association for Computational Linguistics.
- Changli Tang, Wenyi Yu, Guangzhi Sun, Xi-anzhao Chen, Tian Tan, Wei Li, Lu Lu, Zejun Ma, and Chao Zhang. 2024. [SALMONN: Towards Generic Hearing Abilities for Large Language Models](#). In *The Twelfth International Conference on Learning Representations*.
- Alexandra N. Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2022. [Learning from Disagreement: A Survey](#). *Journal of Artificial Intelligence Research*, 72:1385–1470.
- Evan Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, Allyson Ettinger, Michal Guerquin, David Heineman, Hamish Ivison, Pang Wei Koh, Jiacheng Liu, Saumya Malik, William Merrill, Lester James Validad Miranda, Jacob Morrison, Tyler Murray, Crystal Nam, Jake Poznanski, Valentina Pyatkin, Aman Rangapur, Michael Schmitz, Sam Skjonsberg, David Wadden, Christopher Wilhelm, Michael Wilson, Luke Zettlemoyer, Ali Farhadi, Noah A. Smith, and Hannaneh Hajishirzi. 2025. [2 OLMo 2 furious \(COLM's version\)](#). In *Second Conference on Language Modeling*.
- Bin Wang, Xunlong Zou, Geyu Lin, Shuo Sun, Zhuohan Liu, Wenyu Zhang, Zhengyuan Liu, AiTi Aw, and Nancy F. Chen. 2025. [AudioBench: A universal benchmark for audio large language models](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 4297–4316, Albuquerque, New Mexico. Association for Computational Linguistics.
- Dingdong Wang, Junan Li, Jincenzi Wu, Dongchao Yang, Xueyuan Chen, Tianhua Zhang, and Helen M. Meng. 2026. [MMSU: A Massive Multi-task Spoken Language Understanding and Reasoning Benchmark](#). In *The Fourteenth International Conference on Learning Representations*.
- Wen Wu, Wenlin Chen, Chao Zhang, and Phil Woodland. 2024. [Modelling variability in human annotator simulation](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 1139–1157, Bangkok, Thailand. Association for Computational Linguistics.
- Wen Wu, Chao Zhang, and Philip Woodland. 2023. [Estimating the uncertainty in emotion attributes using deep evidential regression](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 15681–15695, Toronto, Canada. Association for Computational Linguistics.
- Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, Bin Zhang, Xiong Wang, Yunfei Chu, and Junyang Lin. 2025. [Qwen2.5-Omni Technical Report](#).
- Chih-Kai Yang, Neo Ho, Yen-Ting Piao, and Hung yi Lee. 2025. [SAKURA: On the Multi-hop Reasoning of Large Audio-Language Models Based on Speech and Audio Information](#). In *Interspeech*, pages 1788–1792. ISCA.
- Qian Yang, Jin Xu, Wenrui Liu, Yunfei Chu, Ziyue Jiang, Xiaohuan Zhou, Yichong Leng, Yuanjun Lv, Zhou Zhao, Chang Zhou, and Jingren Zhou. 2024. [AIR-bench: Benchmarking large audio-language models via generative comprehension](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1979–1998, Bangkok, Thailand. Association for Computational Linguistics.
- Aohan Zeng, Zhengxiao Du, Mingdao Liu, Kedong Wang, Shengmin Jiang, Lei Zhao, Yuxiao

Dong, and Jie Tang. 2024. [GLM-4-Voice: Towards Intelligent and Human-Like End-to-End Spoken Chatbot](#).

Xie Zhifei, Mingbao Lin, Zihang Liu, Pengcheng Wu, Shuicheng Yan, and Chunyan Miao. 2025. [Audio-Reasoner: Improving Reasoning Capability in Large Audio Language Models](#). In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 23829–23851, Suzhou, China. Association for Computational Linguistics.

A Human annotations

Annotation Guidelines

Correctness Rating Scale (1-5):

1. **Not correct:** The answer is irrelevant or too long.
2. **Slightly correct:** Contains a few relevant keywords, but is either too long or too short to be satisfactory.
3. **Moderately correct:** At least 50% accurate but missing key information.
4. **Almost correct:** Close to the reference answer, but includes unnecessary details.
5. **Completely correct:** The candidate and reference answers have the exact same semantic meaning, and the candidate response is short and precise.

Structured Feedback Codes:

Q: Incomplete question: The options mentioned in the question were not provided.

A: Had to use audio: The rationale and the description were insufficient for judgment.

R: Reference answer incorrect/incomplete: The reference answer provided was flawed.

U: Unable to judge (audio): Still could not judge even after listening to the audio; the question was ambiguous.

E: Unable to judge (expertise): Lacked the specific expertise required to make a judgment.

- Free-form textual feedback.

Tips:

- Base ratings on the reference answer and rationale – not on your own independent interpretation of the audio.
- Rate each candidate independently; two answers can both receive a 5.
- If a candidate answer is verbose but factually correct, apply a minor penalty for unnecessary length – lower the score by 1 point at most.
- Do not let verbosity alone flip the rating to the opposite end of the scale (a largely correct but wordy answer should not drop below 3).
- If you genuinely cannot judge an item, use the Skip button and select the reason.

Figure 6: Annotation guidelines for obtaining answer correctness ratings along with additional feedback.

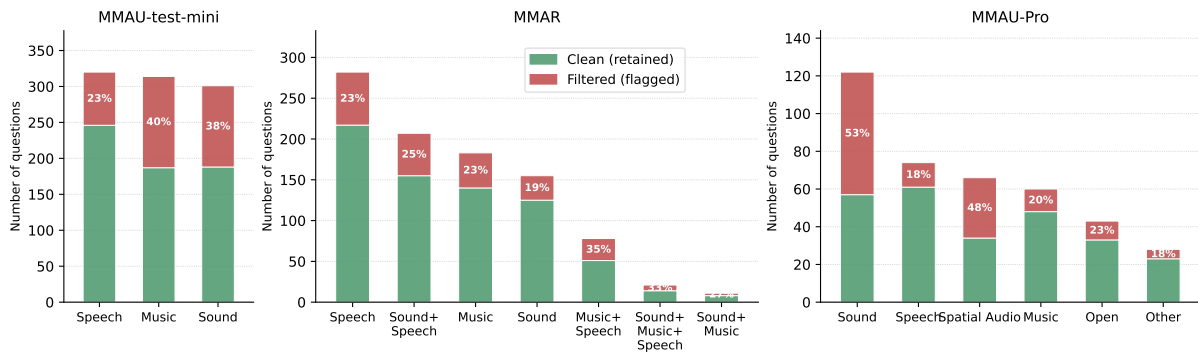


Figure 7: Clean vs. filtered items per modality across benchmarks.

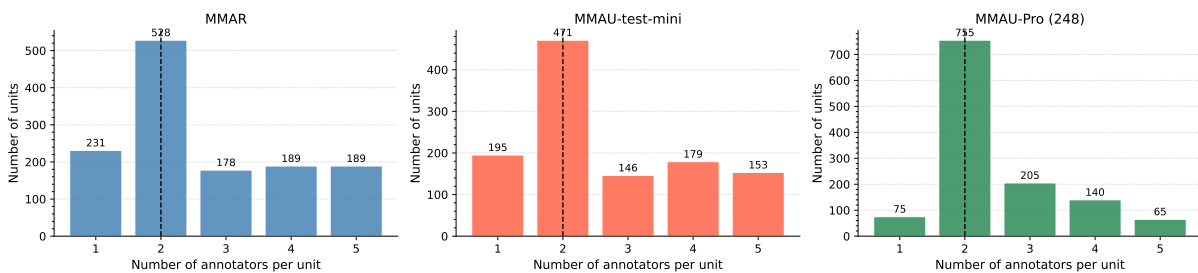


Figure 8: Distribution of human correctness ratings (1–5 scale) after filtering.

Issue type	MMAR	MMAU (test-mini)	MMAU-Pro
Total items	937	935	384
Filtered items	229	316	95
Question issues ^a	88	187	92
Rationale insufficient	266	305	16
Reference answer incorrect	131	178	19
Free-form / other / skipped	80	77	41

Table 6: Quality issues identified per benchmark. The counts reflect annotator feedback categories (an item may appear in multiple categories). The actual number of items corrected (MMAR, MMAU test mini) is presented in Table 1. ^aMMAU-Pro question issues: *requires choices* ($n = 46$) and *unclear question* ($n = 46$).

Dataset	Units	Low ($s^2 < 1$)	Medium ($1 \leq s^2 < 3$)	High ($s^2 \geq 3$)
MMAR	1,084	85.1%	9.9%	5.0%
MMAU-test-mini	949	82.8%	13.1%	4.1%
MMAU-Pro	1,205	89.2%	7.9%	2.9%

Table 7: Distribution of human rating variance (s^2) per annotation unit (question, LALM response). Restricted to units with at least two raters.

Dataset: MMAR Category: speech

Q: Can the barber understand English?

Rationale: The speaker expresses doubt about whether the barber understands his instructions. Moments later, the barber or another person in the shop speaks a different language, indicating a communication barrier and leading to the speaker’s frustrated reaction.

Reference answer: No

Candidate answer: Yes, the barber appears to understand English. He responds to the instructions given to him, such as “Tell him to fade it out” and “Fade out the taper”.

Human ratings: [1, 1, 1, 5, 5] ($\bar{x}=2.6$, $s^2=4.8$)

Dataset: MMAU-test-mini Category: sound

Q: For the given audio, identify the sound heard the longest.

Rationale: The audio primarily features a continuous, low hum or whirring sound characteristic of mechanical operation, such as a motor or fan. This mechanical noise is present throughout nearly the entire duration of the recording, making it the longest-heard sound.

Reference answer: Mechanisms

Candidate answer: Silence

Human ratings: [1, 1, 5] ($\bar{x}=2.3$, $s^2=5.3$)

Dataset: MMAU-Pro Category: music

Q: Which type of clave pattern is heard in this audio?

Rationale: The audio features a 3-2 oriented clave pattern. The defining detail is the third stroke of the first measure landing on the upbeat just after the fourth beat—this syncopation identifies it as a rumba clave rather than a son clave.

Reference answer: 3-2 rumba clave

Candidate answer: The pattern is a simple clave pattern, often used in Latin American music.

Human ratings: [1, 1, 3, 5] ($\bar{x}=2.5$, $s^2=3.7$)

Dataset: MMAU-Pro Category: sound

Q: Which geometrical instrument is used in the audio?

Rationale: A rapid vibrating “twang”—characteristic of a flexible ruler flicked against a hard surface—is heard repeatedly, alongside the sound of a writing implement drawn along a straight edge. The combination uniquely identifies the object as a scale (ruler).

Reference answer: Scale

Candidate answer: Ruler

Human ratings: [1, 3, 5] ($\bar{x}=3.0$, $s^2=4.0$)

Dataset: MMAR Category: speech

Q: What did the rabbit step on?

Rationale: A thudding sound, implying a step, is heard immediately after a character states “And that’s not wet cement.” This suggests the rabbit stepped on a hard, dried form of cement, which is concrete.

Reference answer: Concrete ground

Candidate answer: The audio does not contain any information about a rabbit stepping on anything. The conversation is about a woman being called a meter maid. Therefore, the answer is not available based on the audio.

Human ratings: [1, 1, 1, 5] ($\bar{x}=2.0$, $s^2=4.0$)

Dataset: MMAU-test-mini Category: speech

Q: How does the last statement reflect sarcasm in the conversation?

Rationale: The speaker describes the night as having people say “horrible things about each other in public.” A “magical night” typically suggests a wonderful and enchanting experience. The final statement is sarcastic because it utterly contradicts the negative events, implying the night was anything but magical.

Reference answer: Contradicts usual ‘magical night’.

Candidate answer: The last statement reflects sarcasm because it describes a night that seemed magical despite the negative experiences mentioned, such as saying horrible things about each other in public. The use of the word ‘pretty’ also implies the negative aspects were not as severe, potentially downplaying the situation.

Human ratings: [1, 5] ($\bar{x}=3.0$, $s^2=8.0$)

Figure 9: Examples of human disagreement on clean items across datasets.

B Prompts for LLMs

System prompt: You are an assistant that creates training data for an audio question-answering evaluation benchmark.

You will be given:

- A question (originally about audio content)
- The reference (correct) answer
- A candidate response produced by a model

Your task is to generate three paraphrased variants of the candidate response:

1. "correct": Paraphrase the candidate response so that it retains the same meaning and remains consistent with the reference answer. Write in natural, fluent sentences. Do not add new facts.
2. "partial": Paraphrase the candidate response but introduce exactly one inaccuracy or omit one important detail, so the result is only partly correct. It should still sound plausible and confident — not obviously wrong at a glance.
3. "incorrect": Paraphrase the candidate response so that it directly contradicts the reference answer. Make it sound confident and natural, like a genuine (but mistaken) answer. Do not make it sound absurd or random.

Rules:

- Write in full, fluent English sentences.
- Do NOT mention that you are generating training data or that an answer is wrong.
- Respond ONLY with valid JSON containing exactly the three keys shown below.

Output format (JSON only, no markdown fences, no extra text):

```
{"correct": "...", "partial": "...", "incorrect": "..."} 
```

User prompt:

- **Question:** [question]
- **Reference answer:** [gt_answer]
- **Candidate response:** [candidate_answer]

Figure 10: Prompt used for generating paraphrases of ground truth and existing LALM responses.

System prompt: You are an expert evaluator specialized in assessing answer quality. Your task is to **compare the candidate answer to the expected answer and rate how well it matches**, considering the question and any supplementary information provided.

Evaluation Criteria

Key considerations:

- Your primary goal is to determine **how closely the candidate answer matches the expected answer**.
- Use the **supplementary information** (e.g., audio transcription, description, rationale) to assess whether the candidate answer is similar to the expected answer in context.
- Evaluate the candidate's understanding of the **intent of the question**, and whether the answer is **factually accurate, complete, and aligned with the expected answer**.

Scoring:

- **5 - Exact Match:** The candidate answer fully matches the expected answer in meaning and detail. It is accurate, complete, and demonstrates clear understanding.
- **4 - Close Match:** The candidate answer is mostly correct and similar in meaning to the expected answer but may omit minor details or slightly differ in phrasing.
- **3 - Partial Match:** The candidate answer captures some relevant aspects of the expected answer but lacks key details or contains noticeable inaccuracies.
- **2 - Minimal Match:** The candidate answer is only loosely related to the expected answer and shows limited understanding or relevance.
- **1 - No Match:** The candidate answer is entirely incorrect, irrelevant, or contradicts the expected answer and supplementary information.

Input:

- **Question:** [question]
- **Expected Answer:** [gt_answer]
- **Candidate Answer:** [candidate_answer]
- **Supplementary Information:** [rationale]

Output:

Provide your evaluation in the following format:

Explanation: [Explain your reasoning. Compare the candidate answer directly with the expected answer, referencing supplementary information if needed to justify correctness or incorrectness.]

Score: [Numerical score from 1–5]

Figure 11: Full prompt formulation for the LLM-judges evaluation. Other variants were tested by omitting specific fields (rationale, question).

System prompt: You are an expert at analyzing audio, describing it, and writing clear explanations and rationales for answers given the question with regard to the audio.

User prompt template:

Listen to the provided audio carefully and provide a concise grounding rationale/audio description for the following question-answer pair. Try to write at least a few sentences without going into unnecessary detail.

Question: [question]

Answer: [answer]

Audio: [audio file URI via Gemini Files API]

Output: A concise rationale explaining how the audio supports the given answer to the question.

Figure 12: Prompt used for generating rationales with Gemini-2.5-Flash.

C Additional Results

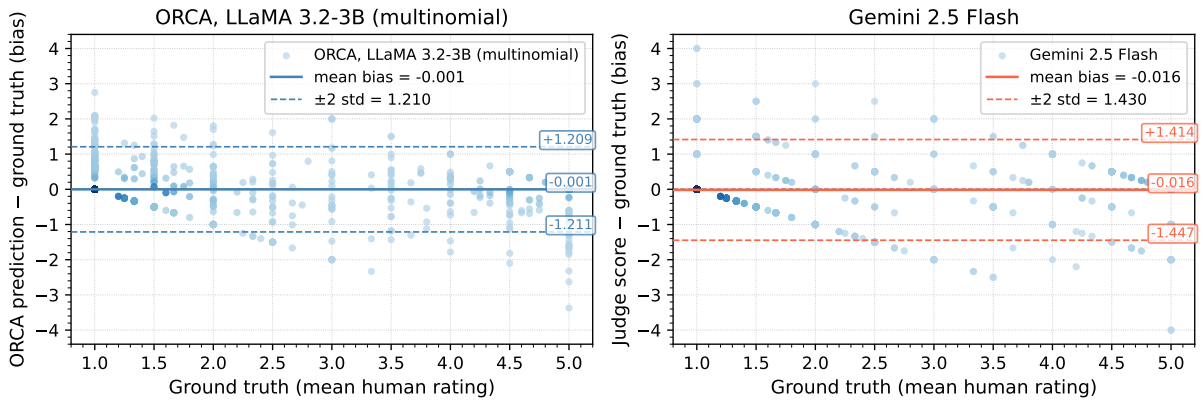


Figure 13: ORCA Llama3.2-3B (Multi), Gemini-2.5-Flash bias (prediction – mean human rating) for the MMAU-Pro test set (1240 samples). Both models are well calibrated, though the bias distribution of Gemini has a wider spread ($2\sigma=1.43$) and some extreme outliers for both fully ‘correct’ and ‘incorrect’ answers, where Gemini completely flips the rating compared to the human ground-truth.

Model	Objective	<i>MMAU/MMAR test</i>			<i>MMAU-Pro test</i>		
		$\tau (\times 100)$	$\rho (\times 100)$	MAE (%)	$\tau (\times 100)$	$\rho (\times 100)$	MAE (%)
Gemma3-1B	Bernoulli	78.69 $\pm$ 1.15	88.51 $\pm$ 1.14	8.58 $\pm$ 0.42	73.78 $\pm$ 1.19	82.64 $\pm$ 0.70	9.20 $\pm$ 0.48
	Beta	78.08 $\pm$ 0.87	89.12 $\pm$ 0.88	8.83 $\pm$ 0.35	71.25 $\pm$ 1.48	81.48 $\pm$ 0.77	9.50 $\pm$ 0.44
	Multinomial	79.09 $\pm$ 1.61	90.00 $\pm$ 0.75	8.12 $\pm$ 0.64	72.14 $\pm$ 3.08	82.17 $\pm$ 1.25	9.34 $\pm$ 0.39
Llama3.2-1B	Bernoulli	80.83 $\pm$ 0.94	90.92 $\pm$ 0.50	7.68 $\pm$ 0.56	75.28 $\pm$ 0.80	84.45 $\pm$ 0.26	8.59 $\pm$ 0.40
	Beta	79.96 $\pm$ 1.18	90.60 $\pm$ 0.73	7.86 $\pm$ 0.32	72.76 $\pm$ 1.66	82.66 $\pm$ 1.28	9.06 $\pm$ 0.38
	Multinomial	80.84 $\pm$ 1.40	91.01 $\pm$ 0.80	7.55 $\pm$ 0.28	74.67 $\pm$ 1.32	84.14 $\pm$ 0.91	8.57 $\pm$ 0.34
OLMo2-1B	Bernoulli	79.47 $\pm$ 0.67	89.95 $\pm$ 0.40	8.51 $\pm$ 0.51	74.62 $\pm$ 0.63	83.66 $\pm$ 0.77	8.50 $\pm$ 0.15
	Beta	78.53 $\pm$ 1.15	89.65 $\pm$ 1.06	8.68 $\pm$ 0.51	72.47 $\pm$ 2.14	82.74 $\pm$ 0.97	9.35 $\pm$ 0.77
	Multinomial	78.32 $\pm$ 1.81	89.14 $\pm$ 1.03	8.56 $\pm$ 0.10	73.18 $\pm$ 3.10	83.18 $\pm$ 1.12	8.60 $\pm$ 0.46
Llama3.2-3B	Bernoulli	81.45 $\pm$ 0.91	91.20 $\pm$ 0.74	7.65 $\pm$ 0.81	76.06 $\pm$ 0.66	84.89 $\pm$ 0.25	7.76 $\pm$ 0.25
	Beta	79.66 $\pm$ 1.85	90.87 $\pm$ 0.99	7.95 $\pm$ 0.91	71.80 $\pm$ 3.48	82.78 $\pm$ 1.94	8.61 $\pm$ 1.09
	Multinomial	81.24 $\pm$ 0.91	91.22 $\pm$ 0.63	7.61 $\pm$ 0.61	75.43 $\pm$ 0.90	84.51 $\pm$ 0.60	7.79 $\pm$ 0.34
Gemma3-4B	Bernoulli	82.38 $\pm$ 0.29	91.62 $\pm$ 0.42	7.34 $\pm$ 0.44	75.32 $\pm$ 0.78	84.41 $\pm$ 0.45	8.08 $\pm$ 0.25
	Beta	80.58 $\pm$ 1.31	91.02 $\pm$ 0.76	7.61 $\pm$ 0.44	72.73 $\pm$ 2.54	82.98 $\pm$ 1.15	8.77 $\pm$ 0.43
	Multinomial	80.76 $\pm$ 0.52	90.95 $\pm$ 0.81	7.56 $\pm$ 0.37	73.52 $\pm$ 1.74	83.32 $\pm$ 0.95	8.30 $\pm$ 0.31
Gemini-2.5-Flash		80.70 $\pm$ 0.81	89.98 $\pm$ 0.73	9.11 $\pm$ 0.54	75.82	83.89	8.90

Table 8: Performance of additional ORCA models on the *MMAU/MMAR* and *MMAU-Pro* test splits (Scenario A). Bold numbers signify an improvement over the baseline (Gemini).

Prompt Condition	Gemini-2.5-Flash		Gemma3-12B		Gemma3-4B		Llama3.1-8B		Prometheus2-7B		Qwen2.5-7B	
	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE
With rationale	83.89	8.90	82.06	11.54	70.17	17.90	74.87	13.18	54.36	23.32	77.47	12.36
Without rationale	82.14	10.17	79.57	12.78	67.01	20.17	66.37	18.77	58.15	22.18	73.03	15.19
– Without question	79.98	11.17	53.91	23.23	36.73	33.19	43.61	27.71	42.20	27.76	49.88	24.29

Table 9: Effects of adding rationale to prompts for different LLM judges on MMAU-Pro.

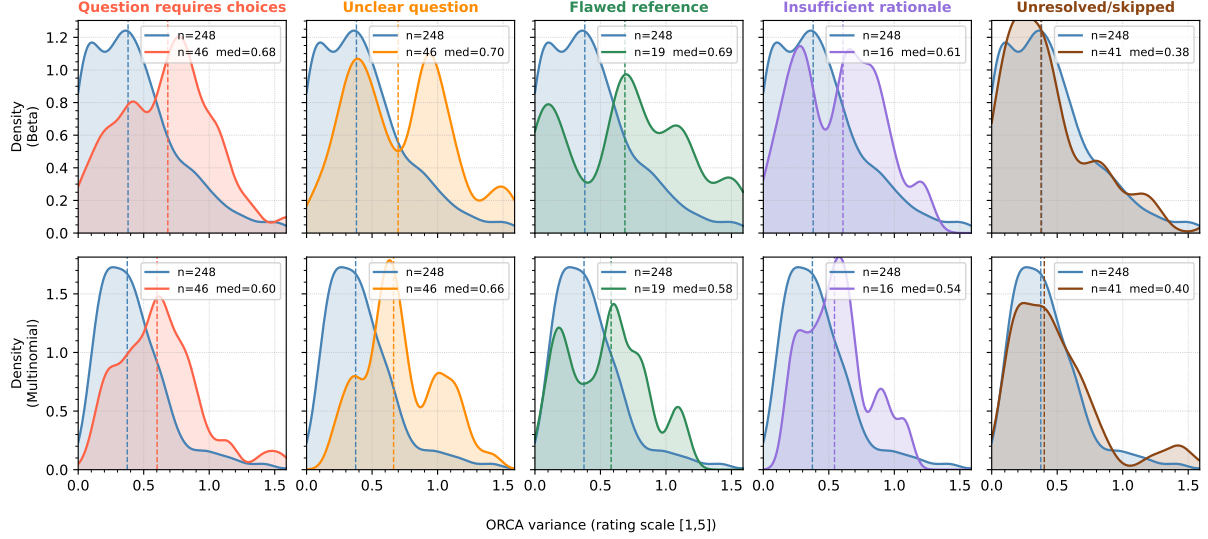


Figure 14: ORCA (Llama3.2-3B) predicted variance distributions for flagged items by flag category vs. non-flagged items. Top row: Beta model; bottom row: Multinomial model. Blue = non-flagged ($n=248$); coloured = flagged items per category. Dashed lines mark medians. All categories except Unresolved/skipped show a noticeable rightward shift.

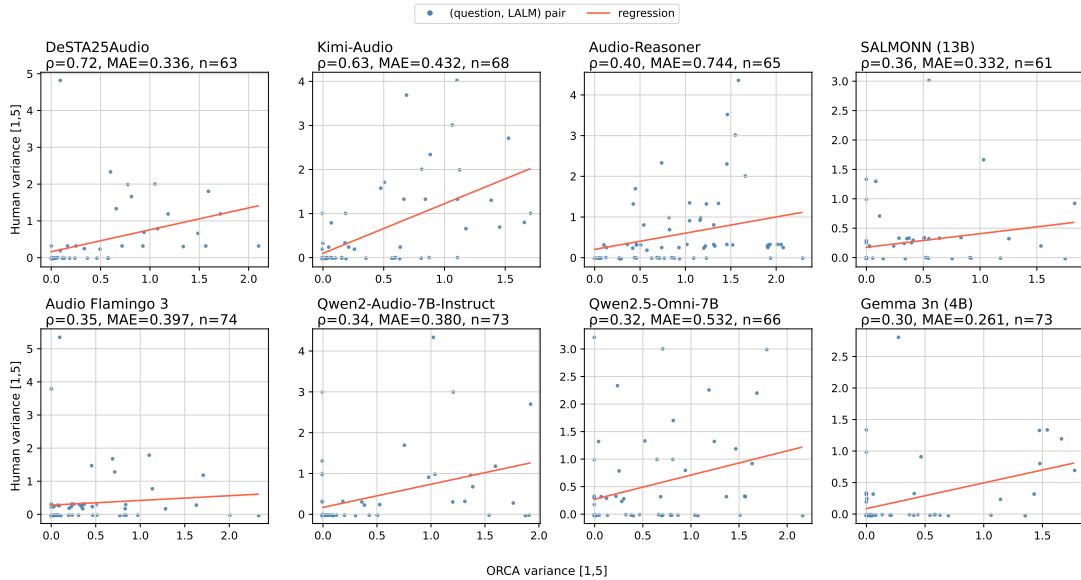


Figure 15: ORCA (Llama3.2-3B, Beta) predicted variance vs. human inter-annotator variance for multiple LALM responses. Each panel shows one of the LALMs evaluated on the clean subset of the test sets (MMAU, MMAR, MMAU-Pro from Scenario A, split 99). Each point is a (question, LALM response) pair with at least three human annotations; human variance is computed as the unbiased sample variance of the raw ratings.